

(1.1) Dokažte, že v libovolném komplexním vektorovém prostoru V se skalárním součinem $\langle \cdot, \cdot \rangle$ platí (pro libovolné $u, v \in V$)

$$\operatorname{Re}(\langle u, v \rangle) = \frac{1}{4}(\|u + v\|^2 - \|u - v\|^2),$$

$$\operatorname{Im}(\langle u, v \rangle) = \frac{1}{4i}(\|u - iv\|^2 - \|u + iv\|^2),$$

kde $\operatorname{Re}(z)$ značí reálnou a $\operatorname{Im}(z)$ imaginární část komplexního čísla z . (Pro inspiraci vizte tvrzení 8.32 ve skriptech.)

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dk 1. tvrzení: (idea důkazu z tvrzení 8.32. ve skriptech)

$$\begin{aligned}\|u + v\|^2 &= \langle u + v, u + v \rangle \\ &= \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle \quad (\text{distributivita})\end{aligned}$$

$$\begin{aligned}\|u - v\|^2 &= \langle u - v, u - v \rangle \\ &= \langle u, u \rangle - \langle u, v \rangle - \langle v, u \rangle + \langle v, v \rangle \quad (\text{distributivita})\end{aligned}$$

$$\begin{aligned}\|u + v\|^2 - \|u - v\|^2 &= (\langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle) - (\langle u, u \rangle - \langle u, v \rangle - \langle v, u \rangle + \langle v, v \rangle) \\ &= 2(\langle u, v \rangle + \langle v, u \rangle) \quad (\text{platí } \langle u, v \rangle + \langle v, u \rangle = 2\operatorname{Re}(\langle u, v \rangle) \text{ podle tvrzení 8.32.}) \\ &= 2(2\operatorname{Re}(\langle u, v \rangle))\end{aligned}$$

$$\begin{aligned}\|u + v\|^2 - \|u - v\|^2 &= 4\operatorname{Re}(\langle u, v \rangle) \\ \operatorname{Re}(\langle u, v \rangle) &= \frac{1}{4}(\|u + v\|^2 - \|u - v\|^2) \quad \blacksquare\end{aligned}$$

dk 2. tvrzení:

$$\begin{aligned}\|u - iv\|^2 &= \langle u - iv, u - iv \rangle \\ &= \langle u, u \rangle + i\langle u, v \rangle - i\langle v, u \rangle + \underbrace{(-i)(\overline{-i})}_{-i \cdot i = -(-1) = 1} \langle v, v \rangle \quad (\text{distributivita}) \\ &= \langle u, u \rangle + i\langle u, v \rangle - i\langle v, u \rangle + \langle v, v \rangle\end{aligned}$$

$$\begin{aligned}\|u + iv\|^2 &= \langle u + iv, u + iv \rangle \\ &= \langle u, u \rangle - i\langle u, v \rangle + i\langle v, u \rangle + \underbrace{(i)(\overline{i})}_{i(-i) = -(-1) = 1} \langle v, v \rangle \quad (\text{distributivita}) \\ &= \langle u, u \rangle - i\langle u, v \rangle + i\langle v, u \rangle + \langle v, v \rangle\end{aligned}$$

$$\begin{aligned}\|u - iv\|^2 - \|u + iv\|^2 &= (\langle u, u \rangle + i\langle u, v \rangle - i\langle v, u \rangle + \langle v, v \rangle) - (\langle u, u \rangle - i\langle u, v \rangle + i\langle v, u \rangle + \langle v, v \rangle) \\ &= -2i(\langle u, v \rangle - \langle v, u \rangle) \quad (\text{platí } \langle u, v \rangle - \langle v, u \rangle = 2i\operatorname{Im}(\langle u, v \rangle)) \\ &= -2i(2i\operatorname{Im}(\langle u, v \rangle))\end{aligned}$$

$$\begin{aligned}\|u - iv\|^2 - \|u + iv\|^2 &= 4\operatorname{Im}(\langle u, v \rangle) \\ \operatorname{Im}(\langle u, v \rangle) &= \frac{1}{4i}(\|u - iv\|^2 - \|u + iv\|^2) \quad \blacksquare\end{aligned}$$